

PEEYUSH KANT MISRA

Data Scientist & Machine Learning Engineer | Anthropic Claude Certified Developer

Email: Pkmisra1999@gmail.com | Phone: +91 9889397735 | Location: India (Open to Global & Remote)
LinkedIn: linkedin.com/in/peeyushkmisra | GitHub: github.com/misrapk | YouTube: @peeyushkmisra

PROFESSIONAL SUMMARY

Results-driven Data Science and Machine Learning professional with 4+ years of hands-on experience in enterprise analytics, data validation automation, and technical education. Anthropic Claude Certified Developer with proven expertise in automating high-volume workflows (reduced data validation errors by 25% at Cognizant), conducting exploratory data analysis at Amazon, and mentoring 400+ students with 750+ published educational ML videos. 1st Place Winner at IIT BHU Technex 2020 Machine Learning Challenge (1st of 150+ teams).

TECHNICAL SKILLS

- Programming Languages:** Python (NumPy, Pandas, Scikit-learn), C++, SQL (PostgreSQL, MySQL), JavaScript, TypeScript
- Machine Learning & AI:** Supervised/Unsupervised Learning, Random Forest, XGBoost, LightGBM, Regression, Neural Networks
- Generative AI & LLMs:** Anthropic Claude (Certified Developer), Claude API & Tool Calling, Prompt Architecture, Gemini, RAG
- Data & Visualization:** Tableau, Matplotlib, Seaborn, Feature Engineering, EDA, Cross-Validation, Anomaly Detection
- Tools & Platforms:** Git/GitHub, AWS (SageMaker concepts), Docker, Linux, REST APIs, Jupyter Notebooks

PROFESSIONAL EXPERIENCE

Cognizant Technology Solutions Sep 2022 – Present | India

Software Engineer (Associate) — Data Validation & Automation

- Engineered end-to-end data validation pipelines and automation rules, slashing verification error rates by 25% across high-throughput enterprise data streams.
- Built automated multi-system reconciliation scripts in Python & SQL, cutting report generation cycle times and improving data reliability for client deliverables.
- Selected for the final round of Cognizant's global Vibe Coding Competition by architecting a full-stack AI application, directly contributing to a Guinness World Records achievement.

Amazon Aug 2021 – Aug 2022 | India

Software Development Engineering (SDE) Intern — Business Analytics & Reporting

- Collaborated with central analytics teams to clean, transform, and analyze massive operational datasets, reducing manual data analysis overhead by over 40%.
- Conducted exploratory data analysis (EDA) and data wrangling utilizing Python (NumPy, Pandas) to extract trends and supply chain optimization insights.
- Built high-impact executive BI dashboards in Tableau to visualize delivery metrics and service level benchmarks for director-level leadership.

Unacademy May 2020 – Nov 2020 | Remote

Coding & Machine Learning Educator (Emerging Educator Award Winner)

- Mentored and instructed 400+ students in Python, C++, Object-Oriented Design, and applied Machine Learning, elevating technical placement readiness.
- Designed and produced 400+ hours of structured video lessons and curriculum, breaking down complex data structures and algorithmic concepts.
- Awarded the Unacademy "Emerging Educator Award" for outstanding instructional clarity, learner satisfaction, and course completion rates.

SELECTED MACHINE LEARNING PROJECTS

Clinical Heart Disease Risk Prediction System [Python, Scikit-learn, Random Forest, XGBoost, Streamlit]

- Engineered a diagnostic model evaluating 14 physiological biomarkers. Conducted multi-collinearity analysis and hyperparameter tuning to achieve 91.4% test accuracy with 0.94 ROC-AUC and calibrated risk probabilities.

FASTag Toll Anomaly & Fraud Detection Classifier [Python, Neural Networks, Pandas, SMOTE, Scikit-learn]

- Built an automated highway toll fraud detection system utilizing temporal transaction velocity features and synthetic minority oversampling. Attained 94.7% accuracy with <0.8% false alarm rate.

Student Academic Performance & Retention Modeling [Python, Multivariate Linear/Ridge Regression, EDA]

- Analyzed socio-demographic and behavioral attributes across 600+ students. Achieved $R^2 = 0.842$ with 1.74 MAE, isolating key dropout indicators for proactive intervention.

CERTIFICATIONS & HONORS

- Claude Certified Developer - Foundations (Anthropic, 2024)
- Claude Certified Associate - Foundations (Anthropic, 2024)
- 1st Prize Winner — Technex 2020 ML Challenge, IIT BHU Varanasi (Outperformed 150+ teams nationwide with 93% model accuracy)

- Cognizant Vibe Coding Finalist & Guinness World Record Hackathon Contributor
- Deep Learning Specialization (Coursera / deeplearning.ai — Andrew Ng)
- AWS Machine Learning (Amazon Web Services)
- HackerRank Gold Badge (5-Star in Problem Solving & Algorithms)
- SQL for Data Science (Coursera / UC Davis) & Python Programming (Univ. of Michigan)

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science and Engineering

2017 – 2021